

MATTHEW H. TAYLOR

🔗 mht3.github.io

Research Focus: Data-driven control, science/physical applications of reinforcement learning and imitation learning.

EDUCATION

UNIVERSITY OF CALIFORNIA, SAN DIEGO

- PhD in Computer Science and Engineering (Est. Graduation: 06/29)
- MS in Computer Science and Engineering

GPA: 4.0 / 4.0

SEP. 2025 - PRESENT

SEP. 2023 - JUNE 2025

UNIVERSITY OF ILLINOIS AT URBANA-CHAMPAIGN

- Bachelor of Science in Aerospace Engineering
- Highest Honors, Minor in Computer Science

GPA: 3.93 / 4.00

AUG. 2019 - MAY 2023

SKILLS

- | | | | |
|----------|---------|------------|--------------|
| • Python | • C/C++ | • Java | • PyTorch |
| • Git | • Unix | • IsaacLab | • TensorFlow |

WORK EXPERIENCE

NASA Jet Propulsion Laboratory, Machine Learning Research Intern JUNE 2024 - NOV. 2024

- Leveraged self-supervised contrastive learning techniques to create an ocean synthetic-aperture radar (SAR) foundation model and finetuned on ocean science object detection and semantic segmentation tasks.
- Developed a versatile, dataset and model-agnostic semantic segmentation and object detection training codebase with integration of Weights and Biases for visualizing and monitoring model performance.
- Curated a dataset with bounding box and polygon mask annotations for over 7,000 submesoscale ocean eddies in the western Mediterranean Sea and prepared for NASA PO.DAAC (NASA oceanography data archive).

Dolby Laboratories, Computer Vision/ML Engineer Intern JUNE 2023 - AUG. 2023

- Profiled video content using skin segmentation, noise, face, and text detection models to give suggestions for when to use precision detail, a local image adjustment algorithm created for Dolby Vision.
- Updated Dolby Vision's internal C++ to Python bridge by adding hooks for local enhancement parameters and produced a modular PyQt GUI to visualize content mapped Dolby Vision content and recommendations.

Maxar Technologies, Research and Development Intern MAY 2022 - AUG. 2022

- Applied deep learning and semantic segmentation methods to detect building footprints from Maxar satellite imagery using PyTorch Lightning for training and TensorBoard for model evaluation and visualizations.
- Researched how the addition of the near infra-red band (NIR) improves segmentation performance.
- Developed model-agnostic visualizations in TensorBoard for the R&D Lab's custom training framework.

RESEARCH

Graduate Researcher, Advised by Professor Sicun Gao MARCH 2024 - PRESENT

- Improving learning-based design for high dimensional control problems using ideas from Lyapunov stability, imitation learning, and learning from imperfect experts.
- Creating more stable, robust policies for the Unitree G1 humanoid robot for balance and locomotion tasks.
- Created an approximate neural Lyapunov controller for pendulum and cartpole gymnasium environments with the goal of expanding and generalizing to problems with larger state spaces and unknown dynamics.

Undergraduate Researcher, Tran Research Group MAY 2021 - MAY 2023

- Created a CNN to detect objects and transform a robot's camera view into a top-down grid world in simulation and the real-world using the NVIDIA JetBot AI kit. Controlled the robot in simulation using Gazebo and ROS.
- Explored useful applications for brain computer interfaces and human-agent teaming, applying machine learning algorithms to EEG data to improve a human-agent collaborative environment.
- Created an experiment with multiple participants and developed a multiclass SVM to classify LEDs flashing at different frequencies based on EEG signals using steady-state visual-evoked potentials (SSVEP).

TEACHING

Teaching Assistant, Intro to Programming (UCSD CSE 8A) JAN. 2024 - MARCH 2024

Mentor TA, Teaching Methods in Computer Science (UCSD CSE 599) SEP. 2024 - DEC. 2024

Teaching Assistant, Search and Optimization (UCSD CSE 257) JAN. 2025 - MARCH 2025

Teaching Assistant, AI Search and Reasoning (UCSD CSE 150B) MARCH 2025 - JUNE 2025

PUBLICATIONS

*Equal Contribution

1. Taylor, M.H., “Advantage Regularization for Imitating Complex Control Policies”, MS Thesis, University of California, San Diego, 2025.

RELEVANT COUSEWORK

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|---------------------------|------------------------------|----------------------------|
| • Search and Optimization | • Random Processes | • Introduction to AI |
| • Algorithms | • Probability and Statistics | • Introduction to Robotics |
| • ML for Physical Appl. | • Parallel Computing | • Machine Learning |

AWARDS

- 2023 Summa Cum Laude - Aerospace Engineering at University of Illinois at Urbana-Champaign
- 2020-2023 James Scholar at University of Illinois at Urbana-Champaign
- 2019 State Farm National Merit Scholar

COMMUNITY SERVICE

Volunteer, Del Mar Dog Rescue

2024 - PRESENT

Del Mar, California

- Walked dogs on weekends to support a local rescue.

Softball Coach, Special Opportunities Available in Recreation

2017 – 2019

Bloomington, Illinois

- Coached children and adults with disabilities in a yearly softball league.

Buddy, The Miracle League of Central Illinois

2017 – 2019

Normal, Illinois

- Helped children and adults with disabilities experience the joy of playing baseball.